

When phase 'Y' is available, LED2 glows. When phase 'B' is available, LED3 glows.

The main advantage of this protection circuit is that it protects three-phase appliances from failure of any of the phases by disconnecting the power supply through the contactor and

automatically restores the three-phase supply to the appliance (with reasonable time delay) when all the phases are available.

Assemble the circuit on a general-purpose PCB and enclose in a cabinet with the relays and contactor mounted on the back side of cabinet.

Connect the appliance through external wires.

Caution. To avoid the risk of electric shock, ensure that AC mains is disconnected during assembly of the circuit and double check everything before connecting your circuit to the mains. ●

MUSCLE STIMULATOR

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Here is a circuit that stimulates nerves of that part of your body where electrodes are attached. It is useful to relieve headache and muscular pain and revive frozen muscles that impair movement. Though it provides muscles stimulation and invigoration, it's mainly an aid in removing cellulitis.

The system comprises two units: muscle stimulator and timer.

Fig. 1 shows the circuit of the muscle stimulator. IC 7555 is wired as an astable multivibrator to generate about 80Hz pulses. The output of IC1

is fed to transistor T1, whose emitter is further connected to the base of transistor T2 through R3 and VR1. The collector of transistor T2 is connected to one end of the secondary winding of transformer X1. The other end of the secondary winding of the transformer is connected to ground.

When IC1 oscillates, transformer X1 is driven by the pulse frequencies generated to produce high voltage at its primary terminals. Separate electrodes are connected to each end of the primary winding of transformer X1. Diode D1 (D1) protects transistor T2 against high-voltage pulses generated by the transformer.

Using potentiometer VR1 you can control the intensity of current sensing at the electrodes. The brightness level of LED1 indicates the amplitude of the pulses. If you want to increase the intensity level, replace the 1.8-kilo-ohm resistor with 5.6 kilo-ohms or higher value up to 10 kilo-ohms.

X1 is a small mains transformer with 220V primary to 12V, 100/150mA secondary. It must be reverse connected, i.e., connect the secondary winding to the collector of T2 and ground, and primary winding to the output electrodes. The output voltage

is about 60V but the output current is so small that there is no threat of electric shock.

Electrodes are made of small, thin-gauge metallic plates measuring about $2.5 \times 2.5 \text{ cm}^2$ in size. Use flexible wires to solder electrodes and connect to the output of the device. Before attaching metal electrodes to the body, wipe them with a damp cloth. After attaching the electrodes to the body (with the help of elastic bands on velcro straps), flip switch S1 to activate the circuit and rotate the knob of intensity-control preset VR1 very slowly until you feel

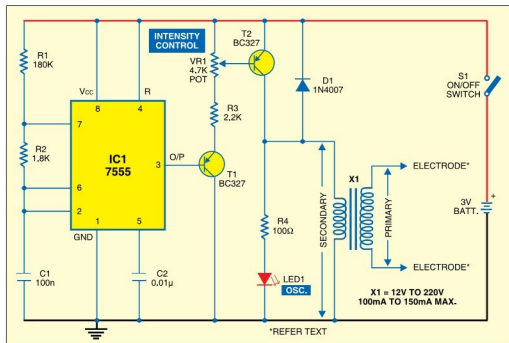


Fig. 1: Muscle stimulator circuit

